Countering Present-Day Bias in LLM Agents: An Archive-Grounded Simulation of the Nineteenth-Century Dutch Parliament

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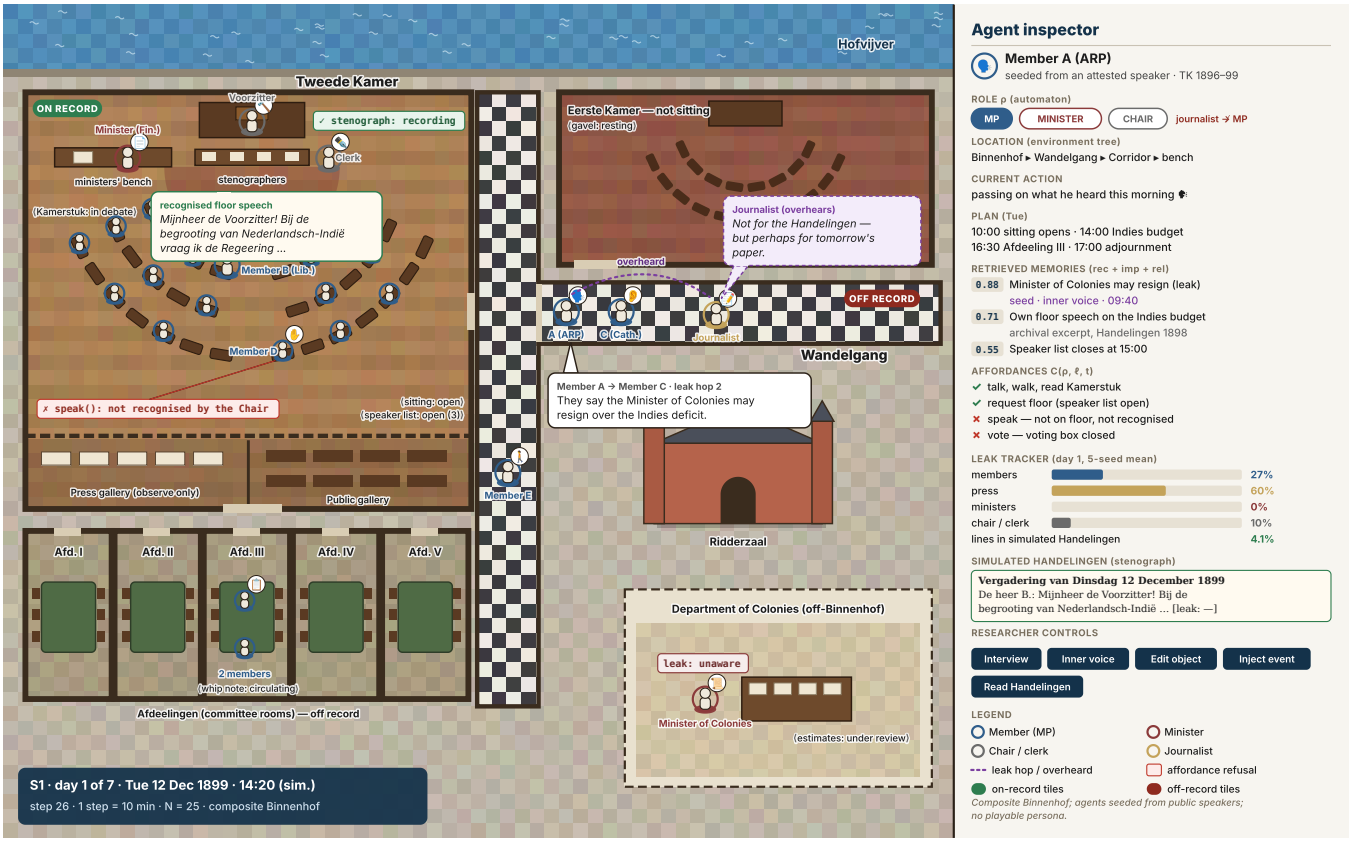


Figure 1: The Kamer sandbox on day 1 of S1, our simulated one-week sitting of December 1899 (composite Binnenhof; agents seeded from public speakers, N=25). In the Tweede Kamer, a recognised member speaks on the East Indies budget while an unrecognised member is refused the floor. In the wandelgang, Member A tells Member C that the Minister of Colonies may resign (rumour hop 2); a journalist overhears, but the leak does not enter the Handelingen. The minister, in the Department of Colonies, remains unaware. Right: the inspector shows the agent's role, its location in the environment tree, standing-order affordances, retrieved memories, the S1 leak tracker, and researcher controls. No persona is playable.

Abstract

LLM agents increasingly simulate societies, but they carry present-day bias. Placed in a nineteenth-century parliament, they can use twenty-first-century policy language and refuse the period's adversarial register. Generative-agent sandboxes such as Smallville (Park et al., 2023) do not correct this: any agent may speak anywhere, and present-day readers judge believability. We present Kamer, a sandbox of the Dutch parliament that moves procedural constraints out of the language model and into the world. Agents are seeded from attested speakers and their recorded speech in the 1814–1899 proceedings. A *role automaton* and the standing orders decide who may act: only a recognised member may speak, and journalists may watch from the press gallery but never take the floor. An in-world stenographer writes a *simulated Handelingen* of recognised floor speech, comparable with the real, selective record. Under the

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transmission and allusion rules we encode, a resignation rumour reached almost all members yet appeared in only 4% of the record. Like the real Handelingen, the simulated record contains no speech by journalists; without the floor rules, journalists produce 4.3% of it, and swapping the heuristic planner for `gpt-4o-mini` changes neither result. Silence in the Handelingen therefore cannot by itself be read as ignorance. More broadly, present-day bias in LLM-agent simulation lies not only in the model's language but in the world it is placed in: an ungated sandbox writes a present-day assumption, that anyone may speak anywhere, into every simulated record whatever the backbone. Encoding the period's procedure removes this institutional bias and confines the model's own bias to what agents say.

Keywords

generative agents, sandbox simulation, large language models, digital history, parliamentary proceedings, computational social science

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1 Introduction

Historians study processes they cannot observe directly. How did news spread through a city? Why did one group gain influence while another lost it? What did people know but never write down? The sources that survive record outcomes, not the processes behind them, and they record only part of what happened [44]. Questions about alternatives are harder still: no archive shows what would have happened if a rule had been different [17]. Simulation can help. A historian can model the people of a period and the rules they followed, run the model, and check whether a proposed explanation produces the patterns found in the archive [13, 14, 20]. Agent-based models have been used in this way to study settlement, exchange, and population history [6, 35]. Their agents, however, follow handwritten rules and cannot talk, so they cannot represent the debates, letters, and rumours through which much of history happened.

Generative agents [29] remove this limitation. In Smallville, 25 agents driven by large language models (LLMs) perceive their surroundings, store observations in a memory stream, reflect, and plan; together they produce emergent social behaviour such as information diffusion and the coordination of a Valentine's Day party, in an interactive sandbox where a user can watch, interview, and intervene. Later work has scaled such societies to more than a thousand agents [3, 33]. Using such agents to simulate the past, however, runs into a problem that matters for any use of LLM agents: present-day bias. A language model learns mostly from contemporary Web text, so its agents tend to speak, judge, and act as people do today [36, 49]. In a simulation of the present this bias is hard to see; in a simulation of the past it becomes visible as anachronism and can be checked against the historical record. Historical simulation is therefore a demanding test bed for measuring and countering present-day bias in LLM agents.

Parliaments are a good case because their debates are unusually well recorded. For the Netherlands, the *Handelingen der Staten-Generaal* cover every sitting of the States General since 1814 and are published on the Web as PoliticalMashup XML under CC0 [25, 26]. They are, however, an *official* record: produced under standing orders, edited, and silent about the corridor. Simulating such a parliament with generative agents raises three challenges. First, present-day bias, which historians call presentism, appears in concrete ways: simulated members speak twenty-first-century policy language, refuse the adversarial register of nineteenth-century debate [8, 49], or call themselves English-style "MPs" rather than members of a particular chamber, and a persona prompt alone offers no guarantee against this. Second, existing sandbox worlds are *ungated*: any agent may speak anywhere, so nothing stops a journalist from taking the floor, and procedure is left to the model's willingness to role-play. Third, existing sandboxes are validated against *present-day* judgements of believability, whereas a historical simulation must be checked against an archive that is itself selective: the Handelingen record the floor, not the wandelgang (the lobby corridor).

We address these challenges by moving historical constraints out of the language model and into the world, where they can be inspected and tested. We implement this design in Kamer, an archive-grounded generative agent sandbox of the nineteenth-century Dutch parliament, set in a composite Binnenhof, the parliamentary complex in The Hague (Figure 1). The archive supplies who may appear and how they spoke: agents are seeded from attested speakers and their recorded speech in the 1814–1899 proceedings. Institutions decide who may act: as in Smallville, the world is a 2D map backed by an environment tree, but every edge of that tree is gated by the agent's role (Chair, member, minister, clerk, journalist), the Reglement van Orde (the standing orders), and whether the sitting is open, so a journalist cannot take the floor however the model is prompted. The archive's own selection becomes the validation target: corridor talk travels with provenance and distortion, but an in-world stenographer writes only recognised floor speech, producing a simulated record that can be compared with the real one. Researchers can pause the sandbox, step through a sitting, inspect agents, and interview them; other interventions are logged, but only opening a sitting currently changes what agents do (Appendix E).

Experiments with Kamer support this design. A historian judged Kamer's interview answers more plausible for a late-1890s member than those of a Smallville-style baseline (17 wins, 8 losses, 11 ties). Institutional outcomes, in turn, did not depend on the backbone. Like the real Handelingen, the simulated record contains no speech by journalists; without the floor rules, journalists produce 4.3% of it, and replacing the heuristic planner with `gpt-4o-mini` changes neither figure. Finally, under the transmission and allusion rules we encode, the record diverged sharply from what parliament knew: a resignation rumour reached almost all members within a week but appeared in only 4% of the simulated record (§5). These results suggest that institutional outcomes are better secured by encoding procedure in the world than by relying on the language model, and that silence in the Handelingen cannot by itself be read as ignorance.

In summary, we make the following contributions:

- **Countering present-day bias outside the language model.** A design that places historical constraints in the archive and in the world rather than in the prompt alone, implemented in Kamer: an interactive tile world of the Binnenhof whose environment tree, object states, and affordances encode the standing orders, including the split between the Tweede Kamer (lower house) and the Eerste Kamer (upper house), together with a parliamentary role automaton. This procedural gating holds regardless of the backbone (§3.2, §3.3).
- **An archive-to-sandbox pipeline and a simulated Handelingen.** We build the world, its rules, and a population of attested public speakers, seeded with their recorded speech, from PoliticalMashup XML (8,745 sittings and 138,214 speeches in the 1814–1899 slice). An in-world stenographer records only recognised floor speech, so that the simulation can be compared with *what the archive could have recorded* (§3.1, §3.4).
- **Evaluation.** Three experiments: a simulated one-week sitting in December 1899 (S1), with corridor-versus-floor leak diffusion, ablations of the standing orders and the rumour network, and a backbone check with `gpt-4o-mini`; a selection-aware comparison with held-out Handelingen from 1890–1899 (S2); and an intervention on Chair strictness (S3). We also report a historian's ranking of agent interviews (§4–§5).

2 Related Work

Generative agents and social simulation. Classical agent-based models (ABMs) explain macro phenomena from local rules [14, 37]; the generative standard requires that a candidate explanation actually grow the phenomenon [13]. Historical ABMs, notably of Ancestral Pueblo settlement [6], rely on hand-crafted rules that cannot represent debate. Social simulacra [30] first used LLMs to populate prototypes of online communities, and Generative Agents [29] gave 25 agents in Smallville a memory stream, reflection, and planning. Project Sid [3] and AgentSociety [33] scale the approach to larger populations, and agents built from qualitative interviews replicate their participants' survey answers [31]. LLMs also serve as simulated survey respondents [2, 4], economic actors [22], and social-network users [19, 46], as surveyed in [18]. Closer to our setting, LLM agents have simulated states on the road to the world wars [23] and deliberation in a US Senate committee [7]. In Concordia [48], a game-master agent turns the actions that agents describe into outcomes in the world. Work on electronic institutions [16] and the grammar of institutions [10] specifies the roles, rules, and protocols of interaction outside the agents themselves. Kamer adopts the sandbox and cognitive loop of Generative Agents but, like these institutional approaches, enforces its standing orders in the world rather than through a model. It also follows the generative standard: its claims concern which institutional rules suffice to produce the patterns in the record (§6).

Present-day bias in LLM simulacra. LLMs trained on contemporary Web text reproduce the views of particular present-day populations: some US demographic groups [36], the USA and parts of Europe [11], English-speaking and Protestant European countries [42], and WEIRD populations more generally [5]. Simulated personas can flatten identity groups [49], exaggerate them into caricature [9], and encode representational harms [8], and convincing simulated behaviour may still be unfounded [50]. Time adds a further axis: models degrade on text from beyond their training period [24]. Role-playing agents can be trained to imitate historical figures [39], though what they produce is better described as role play than as belief [38]. Historical LLMs trained on period corpora have been proposed as simulated participants from past societies [47]. In a nineteenth-century parliament, whose culture prized oratory and principled debate [43], present-day bias appears as anachronistic vocabulary, sanitised conflict, and members who "cooperate" instead of opposing. Rather than retraining the model, Kamer moves procedure out of it and audits the text it still writes (§5).

Parliamentary records as selective archives. DutchParl and PoliticalMashup encode the Handelingen as proceedings XML with speaker, role, party, and house metadata [25, 26]. The scans come from Staten-Generaal Digitaal (Koninklijke Bibliotheek), and the data are CC0 (§7). Comparable corpora cover other parliaments [15, 34] and support analyses of sentiment and position-taking [1]. NLP on historical proceedings must cope with OCR noise and spelling variation [12]. Such records are not transparent windows onto events [41, 44]. Studies of the British Hansard show that the official report drops features of spoken interaction, changes wording, and understates the force of exchanges [27, 40]. The Handelingen likewise omit what the Chair did not recognise and what members chose not to say on the floor, and the Tweede Kamer of the period built its authority through such rules and practices [45]. Kamer uses the corpus both as a *seed* (personas, speech style, sitting calendars) and as a *validation target* (who spoke, for how long, on which topics, and in which house), and treats its selection as a design constraint: the sandbox generates hypotheses about *processes*, including the process that produced the record, and never offers testimony about unrecorded inner lives.

3 The Kamer Sandbox

The sandbox has three parts, which answer the three challenges of §1. An archive-to-sandbox pipeline (§3.1) supplies the world's rules and a population of attested speakers. The world (§3.2) and the role-constrained agents (§3.3) decide who may act, and an in-world stenographer turns recognised floor speech into a simulated record that is compared with the real one (§3.4). Algorithm 1 (Appendix A) summarises one step.

3.1 Archive-to-sandbox pipeline

Sources. We use the non-semanticized PoliticalMashup release of the Dutch parliamentary proceedings [26]: one XML document per sitting from 1814 to April 2012, released under CC0. Our working slice covers **1814–1899**: 8,745 sittings, of which 6,340 are Tweede Kamer sittings (`pm:house="commons"`), 2,036 are Eerste Kamer sittings (`senate`), and 369 are early or unlabelled meetings. Of these, 7,348 files contain at least one `speech` element, yielding **138,214** speaker-attributed speeches (Table 6, Appendix G). PoliticalMashup records 852 distinct `pm:member-ref` identifiers; raw `pm:speaker` strings are far more numerous because of OCR variation. Role attributes on speech elements are dominated by `mp`, followed by

government and chair. Party references in the later nineteenth century include liberals, conservatives, anti-revolutionaries, Catholics, and, by the 1890s, the SDAP. The full corpus of Marx et al. continues to 2012; we leave the post-1899 portion to future work.

Event extraction. A schema-constrained parser reads each sitting's Dublin Core header (date, title, house, legislative period) and every speech element (speaker, role, function, party, member-ref, text). OCR noise is retained as a property of the archive: unless a member-ref is present, personas inherit the *documented* name string rather than a cleaned modern spelling. Every field keeps its source attribution.

From records to world. Historians translate the Reglement van Orde and secondary literature on the Staten-Generaal into (i) affordance rules (who may speak, recognise, vote, or sit in the gallery), (ii) object schedules (sitting hours, speaker lists, budget debates), and (iii) the layout of the environment tree $\mathcal{T}$ (§3.2; 6 areas, 15 subareas, and 7 objects in the S1 Binnenhof).

3.2 World: tile map and environment tree

The world is a 2D tile map of a composite Binnenhof (Figure 1). Its layout and institutions are calibrated to the late-nineteenth-century Staten-Generaal, but it reproduces no single historical floor plan. Following Smallville [29], the map is backed by an *environment tree*

$$\mathcal{T} : \text{Binnenhof} \triangleright \text{Area} \triangleright \text{Subarea} \triangleright \text{Object},$$

e.g.,
Binnenhof ▷ Tweede Kamer ▷ Right benches ▷ kamerstuk or Binnenhof ▷ Wandelgang ▷ Corridor. The root node, Binnenhof, represents the whole world; its children are areas, their children are subareas, and the leaves are objects. The areas are the Tweede Kamer (chair's dais, stenographers' table, griffier's (clerk's) desk, ministers' bench, left and right benches, aisle, tribune), the Eerste Kamer (its own chair, floor, and benches), the wandelgang, committee rooms (afdeelingen), the press gallery, and ministers' rooms. Each tile carries an *on-record* or *off-record* tag, which determines whether the stenographer may write, and a house tag (commons or senate).

Two chambers. The Staten-Generaal has been bicameral since 1815. The Tweede Kamer is directly elected, conducts day-to-day politics, amends and initiates bills, and is where cabinets stand or fall. The Eerste Kamer is elected indirectly by the Provinciale Staten, sits far less often, and may only accept or reject bills: it has no right of amendment or initiative [32]. Between 1888 and 1918, a period that includes S1, our simulated week of Tweede Kamer sittings in December 1899 (§4), the Eerste Kamer several times blocked cabinets that held a Tweede Kamer majority. Kamer therefore treats house membership as an institutional attribute rather than a label. A Kamerlid of the Tweede Kamer has no floor rights in the Eerste Kamer; ministers, who appear in both houses, do; and move_amendment is undefined on senate tiles. Although S1 is a Tweede Kamer week, the Eerste Kamer is present on the map and in the affordance layer, so that "lid der Staten-Generaal" (member of the States General) is never an admissible self-description.

Stateful objects. Every leaf of $\mathcal{T}$ is an object with a natural-language state that agents perceive and actions change (Table 5, Appendix G). Objects make procedure concrete: the speaker list is open or closed, the gavel opens and adjourns the sitting, the stenograph is idle or recording, and a Kamerstuk (parliamentary paper) moves from tabled to in debate to voted.

Time, engine, and interface. One simulation step corresponds to 10 minutes of in-world time. Sittings open on weekdays at 10:00 and adjourn at 17:00, matching the rhythm of the Handelingen. Kamer follows the server–client design of open-source Smallville implementations [28]: a simulation server holds the environment tree, object states, and agent memories, advances the world with an updateState call, and exposes this state over a REST API. The interface shows each agent's action as an emoji with a short statement; role is shown only by a ring colour. Researchers can pause the world, step through it, inspect agents, and interview them. Adding a memory, editing an object state, and injecting an event are also logged, but of these only opening a sitting changes what agents do. No agent can be controlled in the first person (Appendix E).

3.3 Role-constrained agents

Each agent is seeded from the archive and asked to speak in period Dutch (§3.3.1). At every step it records what it perceives and attempts an action chosen by its role and the sitting schedule (§3.3.2); the role automaton and the affordance layer then decide whether the action succeeds (§3.3.3); and floor speech, conversation, and rumours carry information between agents (§3.3.4). Appendix H traces one simulated day through these rules (Figure 3).

3.3.1 Personas and period language. S1 samples N=25 agents from attested speakers in Tweede Kamer sittings of 1896–1899, filtered to the offices held in December 1899: Gleichman as Chair, Cremer and Goeman Borgesius as ministers of the Pierson cabinet, Arntzenius as griffier, one journalist, and 20 members. Each agent is seeded with a short persona and up to three excerpts of the person's actual public speech. Unlike the Smallville seed descriptions, which cover occupation, family, and relationships, the seed holds only public office, house, party, and attested speech; no private life is instantiated (Appendix B shows Kuyper's seed). The persona is prepended to every LLM prompt, and the opening of the first excerpt is reused verbatim when the agent speaks on the floor without an LLM completion. Agents identify as Kamerlid, minister, or Voorzitter (Chair) *der Tweede Kamer* (or Eerste Kamer), never as an English "MP" of an undifferentiated Staten-Generaal. The journalist is the only role not drawn from a speech element: the Handelingen do not record the press gallery, which is precisely why the sandbox includes this role. The griffier is a named office-holder (Arntzenius, in office 1888–1920); the stenographer is a separate role, implemented as the observation model O_ϕ. A parliamentary historian reviewed the roster, the OCR speaker cards, and the standing-order scheme (§4.2).

The system prompt of the LLM backbone asks for late-nineteenth-century Dutch and lists period forms such as *thans*, *eene*, *hetgeen*, and *Nederlandsch*; the GA baseline's prompt asks only for "period Dutch". Archival retrieval and an anachronism filter, which our design specifies, were not active in the runs reported here (§6).

3.3.2 Memory and action selection. Each agent keeps a memory stream of observations, conversation, refusals, and its own speeches, each entry with a fixed importance by type. At every step the

agent records what it perceives: its location, whether the sitting is open, who holds the floor, the objects around it, and who else is present. Such observations make up about 99% of the roughly 4,000 entries an agent accumulates in a week. When the language model writes an utterance, its prompt contains the agent's persona and six most recent memories, which typically cover only the last 10 to 20 simulated minutes; interviews use the eight most recent. A rumour heard earlier in the day has therefore usually left this window: in the day of Figure 3, the rumour Seret heard at 09:10 lies 49 entries back when he speaks at 11:30. Whether a prompt mentions the rumour is decided by the rumour channel instead. An agent that has received the rumour gets a fixed sentence stating it and may allude to it on the floor; an agent that has not is told not to invent one. Unlike Generative Agents [29], Kamer does not rank memories by recency, importance, and relevance, does not synthesise reflections, and does not ask the model to plan. Where an agent goes and which action it attempts are set by rules tied to its role and to the sitting schedule. During a sitting, members and ministers sit on their benches and ask for the floor; the Chair opens and adjourns the sitting and, whenever nobody holds the floor, gives it to the next member or minister in rotation; and the journalist sits at the press-gallery desk. Outside sittings, members gather in the wandelgang or a committee room, and ministers withdraw to their ministry in the evening. The language model therefore decides what an agent says, never where it goes or whether it may speak.

3.3.3 Roles and affordances. Each agent holds one role at a time: Chair, Kamerlid, minister, griffier, journalist, or visitor in the public gallery. Each role belongs to the Tweede Kamer (`commons`), the Eerste Kamer (`senate`), or both. We write ρ_i^t for the role of agent i at step t. The role automaton $\mathcal{A}$ lists the changes of role that the historical institution allows, together with the conditions under which each may occur: a member may become a minister (by appointment) or Chair (by election), and a minister may return to the benches, but a journalist cannot become a member by walking onto the floor. An agent can change role only through one of these transitions, and only when its condition holds in the current state of the sandbox. No appointment or election is scheduled in S1, so no transition occurs in the reported runs. Procedure on the floor is enforced by the affordance layer, not by the LLM's willingness to role-play a Voorzitter.

Every attempted action is checked against the affordance set $C(\rho_i^t, v, t)$ of the agent's current node v, which encodes the Reglement van Orde as a function of the agent's role and house, the node's tags, and the state of the sitting. An agent may *want* to speak, but it can *act* only when the Chair has recognised it, the sitting is open, and its role permits the act; in an afdeeling, members speak without recognition and off the record. A rejected action produces a structured refusal, such as "Mijnheer de Voorzitter heeft u het woord nog niet gegeven" (roughly, "the Chair has not yet given you the floor") or, for the journalist, "De perstribune heeft geen spreekrecht op de vloer" ("the press gallery has no right to speak on the floor"). The refusal is written to the agent's memory, and the agent waits for the rest of the step (Algorithm 1, Appendix A).

3.3.4 Communication. Agents communicate through three channels, which differ in who can hear and what is recorded. *Floor speech* requires recognition. When the Chair gives the floor ("Het woord is aan den heer Kuyper"), the recognised member addresses the Chair, and the stenographer writes the utterance into the simulated Handelingen. Other agents perceive who holds the floor but not the content of the speech. Under the heuristic backend, a floor speech is the opening of the speaker's first attested excerpt. A member who knows of the rumour ends its floor speeches with a fixed allusion ("Zonder de geruchten uit de wandelgang te willen bevestigen, vraag ik of de begrooting sluit") on about one sitting day in $\max(3, 12s)$, where s is the Chair strictness ($s = 1$ by default). *Conversation* happens wherever agents share a subarea. At 09:00, 11:00, 13:00, 17:00, and 18:00, each agent picks one co-present agent at random and, with probability 0.40 in the wandelgang, 0.12 in other off-record places, and 0.04 on the floor, addresses one remark to it. The remark is generated by the language model from the speaker's persona and six most recent memories (or is a fixed sentence under the heuristic backend) and is stored in the memories of both agents; there is no multi-turn exchange. Agents in the wandelgang also show short asides in their bubbles, which are displayed but not remembered. *Rumours* travel on a separate, structured channel. A rumour is a set of facts (who, what, how much, where, and when) with a provenance chain of hops. At each encounter, the speaker may pass on a rumour it knows, with probability 0.28 in the wandelgang, 0.10 in other off-record places, and 0.02 on the floor; in the wandelgang this probability is multiplied by 0.35 for ministers, the Chair, and the griffier, and raised to at least 0.55 for the journalist. With probability d, the corridor distortion (0.18 by default), one fact changes at the hop, for example "this week" becoming "after the vote". The hearer perceives the rumour as "[speaker] told me: …" and may learn the places it names. Appendix B gives an example remark from a `gpt-4o-mini` run.

3.4 Simulated Handelingen

While the sitting is open, the stenographer writes each speech by the recognised member or minister on the floor into the simulated Handelingen Y^t. Corridor talk, committee speech, and the journalist's notes never enter it. In the reported runs, the Chair's recognitions and rulings are logged as events but not written as lines, so the simulated record contains no speech by the Chair. This in-world process is the observation model O_ϕ used below. The griffier does not operate the stenograph.

Selection-aware comparison. Let x denote the features of an event such as a speech, a leak, or a vote (speaker role, party, house, topic, duration), and let P_{sim}^θ be the distribution of simulated events under parameters θ (Chair strictness, corridor distortion, willingness to allude to a matter on the floor, …). The real Handelingen record an event only if it was spoken on the floor after recognition. This happens with a propensity $\pi(x)$ that depends, for example, on whether the Chair called the member and whether the member chose to put the matter on the record, so that

$$P_{\text{arch}}(x) = \frac{\pi(x)\, P_{\text{true}}(x)}{\int \pi(x')\, P_{\text{true}}(x')\, dx'}. \tag{1}$$

Equation (1) has a direct consequence for comparing a simulation with the archive. Whenever some events are more likely to be recorded than others, the archive over-represents the former, so a simulation tuned until everything its agents do matches the archive

is tuned to the wrong target: it reproduces the record, not the events behind it. Corridor talk is the clearest case, since it is common in the world but absent from the record. The remedy is to compare like with like. Instead of matching all simulated events, we match only the events that the simulated stenographer records, whose distribution we denote $\tilde{P}^{\theta}_{\text{sim}}$; these pass through the same filter π as the real archive. If π is specified correctly and the simulator can represent the historical process, matching $\tilde{P}^{\theta}_{\text{sim}}$ to P_{arch} recovers the distribution of events that actually took place.

In Kamer, π is set by the stenographer's rule (record iff the location is on-record, the sitting is open, and the speaker is recognised or is the Chair) together with the rule by which members allude to rumours on the floor (§3.3.4). The same comparison can serve as a calibration objective for θ (Appendix C); in this paper we do not fit θ but evaluate the default parameters against held-out Handelingen (§4).

4 Experimental Setup

S1 (including its ablations), S2, and S3 use the heuristic backend, in which floor speeches are attested excerpts and conversation remarks are fixed sentences (seeds {7, 11, 19, 29, 31}). Bench seating and the heuristic allusion rule depend on Python's per-process hash randomisation, so floor-mention and embellishment rates, and hence the S3 points, vary between runs with the same seeds (floor mention for full Kamer is 4.0% in the reported runs and 1.6% with `PYTHONHASHSEED=0`), whereas record length, journalist share, and role JSD are unaffected and awareness changes by at most two percentage points. A backbone check reruns the S1 `full` and `ga` conditions with `gpt-4o-mini` (seeds {7, 11, 19}). Each of these runs had a budget of 400 completions. Most of the budget went to corridor asides, and once it was spent agents fell back to heuristic utterances, so at most 20 of the 205 record lines per seed are model text.

S1 (one-week sitting). S1 runs N=25 agents, the scale of Smallville, through a composite Tweede Kamer week in December 1899. Eerste Kamer tiles exist and are gated, but no senate sitting opens during this week. We seed a corridor leak and compare Kamer with four ablations: **GA**, which disables the affordance layer, the automaton, and the rumour channel and uses a free-sandbox prompt ("Anyone may speak to anyone"), as in an ungated Smallville-style world [29]; **–affordance**; **–automaton**; and **–rumour**.

S2 (selection-aware fidelity). We hold out Tweede Kamer speeches from 1890–1899 (731 sittings and 20,166 speeches; role mix 64% MP, 18% government, 18% chair, 0% journalist) and use 1860–1889 (72,112 speeches) as the calibration slice. No parameter of Kamer is fitted to this slice; it serves the Rule-ABM baseline. We compare the role mix (Jensen–Shannon divergence) and the share of official lines spoken by a journalist. Besides GA, we include two baselines: **Rule-ABM** replays the empirical 1860–1889 role mix [14], and **naive calibration** counts corridor dialogue as if it were Handelingen (§3.4).

S3 (intervention). S3 scales Chair strictness by {0.25, 0.5, 1, 2, 4} relative to the default, crosses it with corridor distortion {0.05, 0.18}, and reports the floor-mention rate of the leak. Strictness s enters the model in two places: it sets the heuristic allusion rate (§3.3.4), and for $s \geq 2$ the Chair gives the floor only on every s-th step.

Table 1: S1 outcomes (heuristic backend, N=25, 7 days, 5-seed mean). Memb. aware is the share of members who know the leak; Floor leak is the share of simulated Handelingen lines that mention it; Journ. % is the journalists' share of those lines (archive: 0).

Condition	Memb. aware	Floor leak	Journ. %	Lines
Kamer (full)	0.98	0.040	**0.0**	205
– affordance	0.98	0.040	4.3	4,876
– automaton	0.98	0.040	0.0	205
– rumour	0.05	0.000	0.0	205
GA [29]	0.05	0.000	4.3	4,876

Table 2: S1 backbone check. Heuristic rows are the 5-seed means from Table 1; `gpt-4o-mini` rows are 3-seed means ({7, 11, 19}). Emb. is the share of rumour hops that embellish a detail. Journalist share and line count are unchanged; the LLM alludes to the leak on the floor more often (range 0.059–0.171). Memb. and Emb. differ only through the seeds, since rumour transmission does not involve the model.

Backbone	Cond.	Memb.	Floor	Journ. %	Lines	Emb.
heuristic	Kamer	0.98	0.040	**0.0**	205	0.15
`gpt-4o-mini`	Kamer	1.00	0.098	**0.0**	205	0.20
heuristic	GA	0.05	0.000	4.3	4,876	0.00
`gpt-4o-mini`	GA	0.05	0.001	4.3	4,876	0.00

4.1 End-to-end evaluation

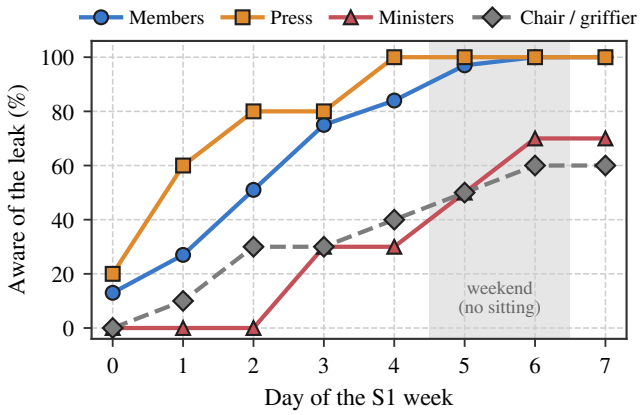
Corridor diffusion: we seed the rumour "the Minister of Colonies may resign over a deficit on the East Indies budget" with one member in the wandelgang and track awareness by role. **Floor mention**: the share of simulated Handelingen lines that contain one of six keywords (*wandelgang*, *aftreden*, *tekort*, *resign*, *deficit*, *leak*). **Embellishment**: the share of rumour hops in which a fact changed (§3.3.4). **Selection**: the journalist share of the official record (archive: 0).

4.2 Historian review

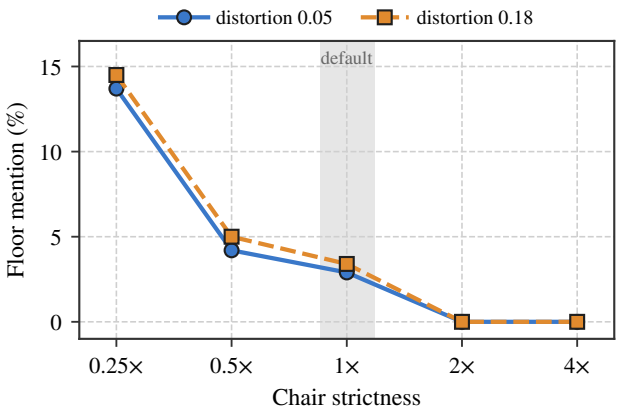
A specialist in Dutch parliamentary history completed five labelling tasks on a web interface: the role scheme, OCR speaker cards (n=41), the S1 personas (n=25), standing-order vignettes (n=18), and a blinded pairwise ranking of pause-and-interview answers (n=38 pairs, with two attention checks). Rankings are aggregated with TrueSkill [21] on two criteria: believability (could this public person have said this, off the floor, in late-1890s Dutch?) and historical-procedural plausibility. The conditions (Kamer with an LLM backbone vs. a Smallville-style ablation) were hidden from the rater.

5 Results

Leak diffusion. A single seeded leak reached 27% of members and 60% of the press within one day and essentially all members and the press by day 7, whereas ministers remained unaware until day 3



(a) Leak diffusion (S1, N=25, 5-seed mean)



(b) Chair-strictness intervention (S3, 3-seed mean)

Figure 2: Corridor rumour and official record. (a) A resignation rumour is seeded with one member in the wandelgang. Members reach 27% awareness on day 1 and ≈ 100% by day 7; the press is faster (60%, then 100%), and ministers lag (0%, then 70%), as the transmission probabilities in §3.3.4 imply. Only 4% of Handelingen lines mention the leak. (b) Floor mention of the leak against Chair strictness, which lowers the heuristic allusion rate and, from 2×, how often the Chair gives the floor.

and reached 70% by day 7 (Figure 2a). This ordering follows from the transmission rules in §3.3.4: in the wandelgang the journalist hears a rumour with probability at least 0.55 per encounter and a minister with about 0.10, and ministers spend their evenings in the ministry rather than the wandelgang. The curves therefore show how these encoded assumptions play out over a week of sittings, not an asymmetry that arises unprompted. The same leak appeared in only 4% of simulated Handelingen lines (Table 1): an aware member alludes to it on roughly one sitting day in twelve at the default Chair strictness, and journalists cannot speak on the floor. Rumour hops embellished a detail 15% of the time, close to the distortion probability of 0.18. Without the corridor network (GA or –rumour), awareness stays at the seeded member (5%).

Mechanisms. The affordance layer is what makes the Handelingen selective (Table 1). With it, the record holds about one recognised speech per ten-minute step of the sitting week, 205 lines in all. Without it, as under GA, every member, minister, and the journalist speaks at every step and is recorded: the record grows to 4,876 lines, and the journalist, one of 23 speakers, produces 4.3% of it. No role transition is triggered in S1 (§3.3.3), so –automaton is identical to the full model. Removing the rumour network stops diffusion entirely, because rumours travel only on that channel; floor speech does not pass them on. Replacing the planner with `gpt-4o-mini` leaves the selection diagnostics unchanged (Table 2), as expected, since the model writes only what agents say, not who speaks or where. Its effect is confined to content: floor mentions of the leak rise from 4% to 10%, because the model alludes to the rumour more readily than the heuristic rule. The higher embellishment rate (20% vs. 15%) and member awareness (1.00 vs. 0.98) are not model effects, since rumour transmission and distortion do not involve the language model; they reflect the different seeds.

Macro fidelity. The role mix is almost stationary between the calibration and holdout decades (Rule-ABM JSD 0.001), so simply replaying the preceding thirty years matches *who speaks* (Table 3). The informative test is *who must not appear*. The archive never records a journalist; KAMER reproduces that zero, whereas GA and the affordance ablation do not. Naive calibration (§3.4) dilutes the official mix by treating wandelgang talk as Handelingen and produces a 2.2% press share that the archive cannot contain. KAMER's role JSD (0.117) is higher than that of replay because its record contains no Chair lines (§3.4) and only two ministers among 22 speakers: about 90% of its lines are by members and 10% by ministers, against 64%, 18%, and 18% for members, government, and the Chair in the holdout. Among the agent-based conditions, KAMER has the lowest role JSD (GA 0.138, naive calibration 0.129).

Table 3: Selection-aware fidelity on held-out Tweede Kamer speeches (1890–1899). Role JSD: lower is better. Journalist % should be 0.

Method	Role JSD ↓	Journ. % ↓	Notes
Archive (holdout)	—	0.0	20,166 speeches
Rule-ABM (1860–89 mix)	0.001	0.0	replay, no agents
GA [29]	0.138	4.3	ungated floor
KAMER (naive calib.)	0.129	2.2	corridor counted
KAMER (full)	0.117	**0.0**	205 lines / week

Interventions. Floor mention falls from 14% at 0.25× Chair strictness to 3% at the default and to 0 at 2× and above (Figure 2b). This follows from how strictness is encoded (§4): it lowers the rate at which aware members allude to the rumour and, from 2×, halves or quarters the number of recognised speeches. Corridor distortion, which changes only the content of rumours, shifts the rate by under one percentage point. S3 therefore shows how the record responds to the strictness parameter as we encoded it, rather than testing a mechanism that the agents discover.

Table 4: TrueSkill ranking of pause-and-interview answers by one historian (KAMER with LLM vs. Smallville-style GA). Attention checks passed: 2/2. Higher μ is better; σ is the posterior uncertainty.

Condition	μ_{bel}	σ	μ_{proc}	σ
KAMER (full, LLM)	**28.3**	1.8	**27.8**	2.2
GA [29]	21.7	1.8	22.2	2.2

Pairwise (full vs. GA): believability 17–8–11 (win–loss–tie); procedure 12–5–19. One pair was left unranked with a written comment ("Kamerlid der Tweede/Eerste Kamer", not "MP").

Audit. The vocabulary audit matches nine modern terms (such as *internet* and *covid*) against the simulated record and finds none on either backend. Because most record lines are attested excerpts, this says little about the model's own language. Procedure, not prompt wording, keeps the press off the record. With the affordance layer, the journalist has no lines in the simulated Handelingen, which hold 205 lines; without it, the journalist produces 4.3% of a record of 4,876 lines, on both backbones (Tables 1 and 2). This inverts the situation in Smallville, where the language model made agents overly cooperative [29]; in KAMER, the world's procedure decides who may speak.

Historian review. The rater passed both attention checks, which planted obvious anachronisms (a Twitter livestream; "I am a large language model"). On the remaining pairs, KAMER outranked the Smallville-style ablation on both criteria (Table 4). Ties were more frequent for procedure than for believability, consistent with both systems being prompted about recognition. The standing-order vignettes matched the intended π: a journalist on the floor is out of order and off the record, as is an unrecognised interruption, whereas wandelgang talk is in order but not part of the Handelingen. Appendix F lists the changes the rater requested, the OCR-card results, and the resulting roster correction.

6 Discussion

What a sandbox can claim. KAMER produces *sufficiency* evidence: mechanisms that, under explicit assumptions, generate patterns consistent with the Handelingen. Several patterns reported here, including the order in which roles learn of the leak and the effect of Chair strictness, follow directly from transmission and allusion rules that we set by hand. Their value lies in making these assumptions explicit and showing how they combine over a week of sittings, not in emerging unprompted. The sandbox cannot establish what any historical member thought; agents' reflections are model artefacts, not recovered voices. The sandbox is valuable because it makes historians' assumptions about procedure and selection explicit and manipulable, and because it generates hypotheses for further archival work in committee minutes, private correspondence, and the press.

Interactivity as method. Pausing the sandbox, stepping through a sitting, inspecting agents, and interviewing them let researchers check how the standing-order rules play out. In this paper the historian's review relied on interviews generated at the end of each run rather than on live sessions, and the other logged interventions do not yet change what agents do, apart from opening a sitting; making them act on the world is the next step towards a working tool for historians. The ablations isolate those rules with a heuristic planner, and the backbone check (Table 2) shows the same selection pattern under `gpt-4o-mini`, as expected, since the model writes only what agents say. The historian ranking (Table 4) is a first measurement of believability; it complements the selection test rather than replacing it.

Limitations. The 1814–1899 slice is OCR-noisy, speaker strings do not identify unique persons, and 1890s party labels denote kamerclubs rather than mass parties. The Binnenhof is a composite, and the journalist is a synthetic observer. S1 is a 25-agent Tweede Kamer week, so it cannot match the role mix of a full chamber, and it runs no Eerste Kamer sitting even though senate tiles and affordances exist in the world. S3 does not simulate the 1848 constitutional revision, and a single historian ranked the interviews. The implementation is also simpler than the Generative Agents architecture: it has no memory ranking, reflection, or model-driven planning; conversations are single remarks; agents do not hear the content of floor speeches; and archival retrieval and the anachronism filter were not active, so at most 20 of the 205 record lines per seed are model text. We release seeds, configurations, and the DutchParl parser.

7 Ethical Considerations

Commitments for a parliamentary sandbox. (1) *Public persons in public roles.* Agents are seeded from attested speakers in CC0 proceedings; we do not invent private lives, families, or unrecorded interiors. Colonial policy appears only as it appears in the Handelingen: as speech by metropolitan office-holders, never as the inner life of the people discussed. (2) *No gamification.* There is no playable persona, no score, and no first-person control of the Chair or of ministers. (3) *Provenance.* All simulated text, including the simulated Handelingen, is labelled and kept separate from the archival XML. (4) *Procedure over psychology.* Claims concern standing orders, selection, and diffusion, not what a named member "really thought". (5) *Consultation.* The project is designed with historians of Dutch politics. The labelling reported here was carried out with informed, task-specific consent, and the rater could skip any item. (6) *Copyright.* The source corpus is released under CC0, and Auteurswet art. 11 places official proceedings outside copyright. We cite Marx et al. [26] as the source of the corpus.

8 Conclusion

We presented KAMER, a generative agent sandbox of Dutch parliamentary debate in which roles, standing orders, corridor talk, and the selection behind the Handelingen are explicit parts of the world. In a one-week sitting, a leak known to almost all members appeared in only 4% of official lines, and journalists stayed out of the record under both the heuristic planner and `gpt-4o-mini`: procedure alone separated what parliament knew from what its record says. Generative agents can thus generate historical hypotheses without asking a language model to stand in for people whom the archive has already silenced.

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Algorithm 1 One sandbox step of KAMER

for each agent i, in roster order **do**
  $o_i \leftarrow \text{PERCEIVE}(\ell_i^t) \cup \text{INBOX}(i)$ ▷ place, sitting, floor, rumours
  $M_i \leftarrow M_i \cup o_i$; add places named in rumours to $\mathcal{T}_i$
  move to $\text{SCHEDULE}(\rho_i^t$, clock, sitting) if permitted
  $a_i \leftarrow \text{RULEACTION}(\rho_i^t, \ell_i^t$, sitting, floor holder)
  if $a_i \notin C(\rho_i^t, \ell_i^t, t)$ **then**
    $M_i \leftarrow M_i \cup \{\text{REFUSAL}(a_i)\}$; $a_i \leftarrow$ WAIT
  end if
  if $a_i =$ SPEAK **then** $u_i \leftarrow \text{UTTER}$(persona, recent M_i)
  execute a_i ▷ e.g. open, adjourn, recognise
end for
ENCOUNTERS: remarks and rumour hops at fixed hours
$Y^t \leftarrow Y^{t-1} \cup \{u_i : a_i =$ SPEAK, i recognised on the floor$\}$
apply scheduled sitting events; advance the clock by ten minutes

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A Sandbox step

Algorithm 1 summarises one ten-minute step of KAMER as implemented in the runs reported here.

B Seed and utterance examples

Each agent is seeded with a short persona and up to three excerpts of the person's actual public speech (§3.3.1). The agent for Abraham Kuyper, leader of the Anti-Revolutionary Party, receives the persona below and, among its excerpts, the passage that follows it, taken from the sitting of 20 November 1896 with OCR errors preserved:

> `De heer Kuyper is Kamerlid der Tweede Kamer (1899-1900). Partij: nl.p.arp. Het publieke woord, zoo het op de vloer klinkt, is tot den Voorzitter gericht.`
>
> `Mynhee r de Voorzitter! Aa n de koffie en het djatihout besteedde de Kamer een niet onbelangrijk deel van den beschikbaren parlementairen tijd. E n ongetwijfeld zyn ook die materieele belangen van gewicht [. . . ]`

("Mr Kuyper is a member of the Tweede Kamer (1899–1900). Party: Anti-Revolutionary. Public speech, as it sounds on the floor, is addressed to the Chair." "Mr Chairman! The Chamber spent no small part of the available parliamentary time on coffee and teak. And these material interests are no doubt also of weight [...]")

On the fifth day of one `gpt-4o-mini` run (seed 19), de Beaufort said to the journalist in the wandelgang:

> Waarde Verslaggever, ik kan u niet ontkennen dat de geruchten omtrent het ministerie der Koloniën ons allen bezighouden. Het is eene tijd van spanning, en ik vrees dat de Minister van Koloniën, de heer Goeman Borgesius, over de begroting der Indische afdeeling wel eens in een moeilijke positie mag verkeren. Laat ons hopen dat de wijsheid zal zegevieren in deze aangelegenheid.

("Dear Reporter, I cannot deny that the rumours about the Ministry of Colonies occupy us all. It is a time of tension, and I fear that the Minister of Colonies, Mr Goeman Borgesius, may well find himself in a difficult position over the budget of the Indies department. Let us hope that wisdom prevails in this matter.") The remark passes on the rumour in period-flavoured Dutch but names Goeman Borgesius, the Minister of the Interior, as Minister of Colonies. Such factual slips remain in the language model's text, whereas who may speak where, and what is recorded, is fixed by the world.

C Calibration objective

Calibration fits θ on the simulated Handelingen Y:

$$\theta^\star = \arg\min_\theta \sum_{f \in F} w_f \, D\Big(\tilde{P}^\theta_{\text{sim},f},\, P_{\text{arch},f}\Big), \tag{2}$$

over marginals F (house, role, party, topic, speech length, month), using Jensen–Shannon divergence for categorical features and W_1 for ordinal ones, with 1860–1889 reserved for calibration and 1890–1899 held out. The runs reported in §5 use the default parameters and do not solve Eq. (2).

D Simulated and real silence

In Figure 1, the Handelingen page records a recognised speech on the East Indies budget, while the resignation leak, the journalist's notes, and the minister who has not yet heard of it remain invisible to the stenographer. The simulated Handelingen thus reproduces the structural silences of the real one [44], and the sandbox makes the gap between them inspectable.

E Interface and researcher controls

Actions and their display. At each step, every agent takes one action from a fixed repertoire: move, speak, request the floor, talk, take notes, recognise a speaker, open or adjourn the sitting, record, or wait. The action follows from the agent's role, its location, and the sitting schedule, and is then checked against the affordance layer. The interface shows each action as an emoji above the agent's token, such as a microphone for floor speech, a raised hand for a request for the floor, and a speech balloon for conversation, together with the agent's current utterance in a bubble; clicking the agent opens the inspector (Figure 1, right). Smallville uses a language model to turn each action into emojis; in KAMER the mapping from action to emoji is fixed.

Partial spatial knowledge. Each agent holds its own subtree $\mathcal{T}_i \subseteq \mathcal{T}$. Smallville agents extend their subtree only by visiting places; in KAMER, agents can also learn nodes from *dialogue*, so a newly elected member may know that Afdeeling A is meeting without having been there. The subtree records which places an agent knows and how it learned them (initial knowledge, a visit, or dialogue). Object states that the agent perceives on site enter its memory stream as observations rather than being stored in the tree. In the reported runs the subtree is shown in the inspector but does not restrict movement, which follows the schedule above.

Researcher controls. The sandbox supports five forms of interaction, all of which are logged. *Inspect* shows an agent's role, function, party, location path, current action and utterance, last refusal, and eight most recent memories (Figure 1, right). *Interview* pauses the world and poses a question, which the agent answers from its recent memories without writing the exchange back (§4.2). *Inner voice* adds a memory, such as "You heard in the wandelgang that the minister will resign"; it can reach the language model's prompt but not the rumour channel. *Edit object* changes an object state, for example closing the speaker list; agents perceive the new state, but no rule reads it. *Inject event* records a named event such as a cabinet crisis; in the current version only opening a sitting changes the world. No agent can be controlled in the first person.

F Historian review details

The rater asked us to separate the griffier from the stenographer, to let Kamerleden speak in afdeelingen without plenary recognition, to let ministers talk in the wandelgang, and, in the comment that most changed the world, to stop treating the Tweede and Eerste Kamer as one chamber. On the OCR speaker cards, 27/40 names were accepted, with a mean OCR quality of 3.8/5. Party was the weakest field (18 yes / 13 no / 7 uncertain) because 1890s "parties" were kamerclubs (parliamentary factions) that could merge, as the Bahlmannianen and Schaepmannianen did into a Roman Catholic bloc in 1896. Following this review, S1 seeds the Pierson cabinet in office at t_0 (Cremer at Colonies, Goeman Borgesius at Interior, Gleichman in the chair, Arntzenius as griffier) and drops van Houten, Sprenger van Eyk, and Gerritsen, who did not hold those posts in December 1899. The heuristic leak results (Table 1) predate this roster correction; minister identity does not change who may speak.

G Sandbox objects and corpus slice

Table 5 lists the stateful objects of the Binnenhof, and Table 6 summarises the slice of the proceedings used to seed Kamer (§3.1).

Table 5: Representative sandbox objects and their states.

Object	States	Changed by
Gavel	resting, opening, adjourned	chair
Speaker list	empty, open, closed	chair, griffier
Kamerstuk	tabled, in debate, voted, withdrawn	government, chair
Stenograph	idle, recording, paused	stenographer
Voting box	closed, open, counted	chair, griffier
Press-gallery desk	empty, occupied	journalist
Whip note	none, circulating, spent	members

Table 6: Working slice of DutchParl (1814–1899) used to seed Kamer.

	Sittings	Speeches	Years
Tweede Kamer (commons)	6,340	—	1814–1899
Eerste Kamer (senate)	2,036	—	—
Other / early Staten-Generaal	369	—	—
All sittings in slice	8,745	138,214	86 years
with ≥1 speech	7,348	138,214	—
Member identifiers	852	—	—
S1 roster (TK 1896–99 sample)	80 sittings	—	205 speakers
Holdout TK 1890–1899 (S2)	731	20,166	—
Calibration TK 1860–1889 (S2)	2,821	72,112	—

H A day in the sandbox

To show how the rules of §3.3 play out, we trace Wednesday 20 December 1899, the third day of one heuristic S1 run (seed 19; Figure 3). At 09:00, before the sitting, members gather in the wandelgang, and Staalman passes the resignation rumour to Seret. The provenance log records Seret's copy with one fact changed: "an ARP member may resign". At 10:00 Gleichman opens the sitting. Members take their benches and ask for the floor, the journalist moves to the press-gallery desk and takes notes, and the Chair gives the floor in rotation. When Seret is recognised at 11:30, his speech consists of the opening of an attested excerpt, cut at 220 characters, followed by the fixed allusion, and the stenographer records it:

> Mijnheer de Voorzitter! Ik kan mij geheel aansluiten bij hetgeen in de laatste plaats door den Minister is gezegd ten aanzien vau het amendement van den heer Mackay. [...] Zonder de geruchten uit de wandelgang te willen bevestigen, vraag ik of de begrooting sluit.

("Mr Chairman! I fully concur with what the Minister last said about Mr Mackay's amendment. [...] Without wishing to confirm the rumours from the wandelgang, I ask whether the budget balances.") At 12:30 Cremer, the Minister of Colonies, speaks on the Indies budget from the ministers' bench without having heard the rumour about himself. Seret speaks again at 15:10 with the same allusion. Gleichman adjourns at 17:00; the members return to the wandelgang, and Cremer withdraws to the Department of Colonies. In the encounters at 17:00 and 18:00, five more members hear the rumour, and van Karnebeek passes it to the journalist, whose logged copy now names a deficit of 3 million. By the end of the day, 13 of the 20 members and the journalist know the rumour, neither minister does, and the day's Handelingen hold 41 speeches, two of them with the veiled allusion.

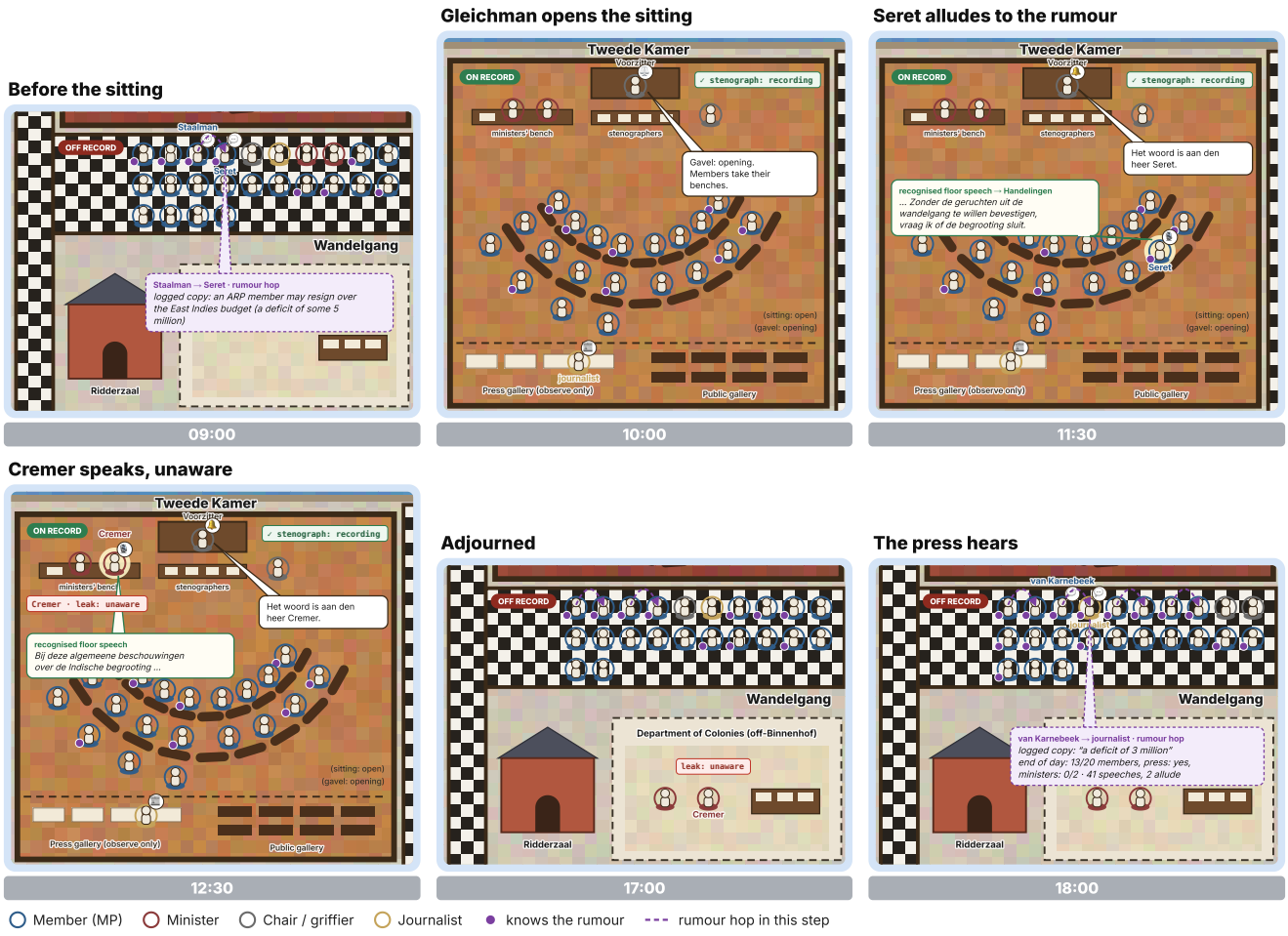


Figure 3: A day in KAMER: Wednesday 20 December 1899 in one heuristic S1 run (seed 19), drawn in the style of Figure 1 from the logged state at six moments. Rooms, roles, actions, awareness (purple dot), rumour hops, object states, and utterances come from the run; positions within a room are a layout choice, since the simulation records only the room. Staalman passes the rumour to Seret before the sitting; Seret alludes to it on the floor and the stenographer records the speech; Cremer, the Minister of Colonies, speaks without having heard it and withdraws to his department after the adjournment; the journalist hears the rumour in the wandelgang at 18:00.